

PART 3 - GRADIENT DESCENT CALCULATION.

$$\text{Model: } y = m_1 x_1 + m_2 x_2 + b$$

Given:

$$X = \begin{bmatrix} 1, 3 \\ 4, 10 \end{bmatrix} \quad \text{targets } [5, 6]$$

$$m = [-1, 2], \quad b = 1, \quad \alpha = 0.01, \quad n = 2$$

Gradient of the MSE

$$J = (1/n) \|y - \text{target}\|^2, \quad \text{let } e = y - \text{target} = X m + b - \text{target}$$

$$\partial J / \partial m = (2/n) \cdot X^T e \quad n = 2 \rightarrow 2/n = 1, \text{ so}$$

$$\partial J / \partial b = (2/n) \cdot \sum e_i \quad \nabla_m = X^T e, \quad \nabla_b = \sum e_i$$

Iteration 1 ($m = [-1, 2], b = 1$)

Predict:

$$y_1 = (-1)(1) + (2)(3) + 1 = -1 + 6 + 1 = 6$$

$$y_2 = (-1)(4) + (2)(10) + 1 = -4 + 20 + 1 = 17$$

$$y = [6, 17]$$

Error:

$$e = y - \text{target} = [6 - 5, 17 - 6] = [1, 11]$$

Gradient:

$$\nabla_m = X^T e = \begin{bmatrix} 1, 4 \\ 3, 10 \end{bmatrix} \cdot [1, 11]$$

$$= [(1)(1) + (4)(11), (3)(1) + (10)(11)] = [1 + 44, 3 + 110]$$

$$= [45, 113]$$

$$\nabla_b = 1 + 11 = 12$$

Update:

$$m_1 = [-1, 2] - 0.01 [45, 113] = [-1.45, 0.87]$$

$$b_1 = 1 - 0.01(12) = 0.88$$

$$m = [-1.45, 0.87] \quad b = 0.88$$

From the first iteration

$$m = [-1.45, 0.87] \quad b = 0.88 \quad x = [(1, 3), (4, 10)]$$

$$y = m_1 x_1 + m_2 x_2 + b \quad y = (5, 6)$$

$$y_1 = (-1.45)(1) + (0.87)(3) + 0.88$$

$$= -1.45 + 2.61 + 0.88 = 2.04$$

$$y_2 = (-1.45)(4) + (0.87)(10) + 0.88$$

$$= -5.8 + 8.7 + 0.88 = 3.78$$

$$\therefore y = [2.04, 3.78]$$

$$\text{Error : } 2.04 - 5 = -2.96 \quad 3.78 - 6 = -2.22$$

$$= [-2.96, -2.22]$$

$$\text{Gradient : } X_c^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix} = \begin{matrix} (1)(-2.96) + (4)(-2.22) \\ -11.84 \\ (3)(-2.96) + (10)(-2.22) \\ -31.08 \end{matrix}$$

$$V_m = [-11.84, -31.08]$$

$$V_b = -2.96 + (-2.22) = -5.18$$

$$\text{Update : } m_1 = -1.45 - 0.01(-11.84) = -1.3316$$

$$m_2 = 0.87 - 0.01(-31.08) = 1.1808$$

$$b_{\text{new}} = 0.88 - 0.01(-5.18) \\ = 0.9318$$

$$m = [-1.3316, 1.1808]$$

$$b = 0.9318$$

$$y = w_1 x_1 + w_2 x_2 + b$$

$$x = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

$$w = [-1.3316, 1.1808], \quad b = 0.9318$$

Predict

$$\hat{y}_1 = (-1.3316)(1) + (1.1808)(3) + 0.9318$$

$$= -1.3316 + 3.5424 + 0.9318 = \underline{3.1426}$$

$$\hat{y}_2 = (-1.3316)(4) + (1.1808)(10) + 0.9318$$

$$= -5.3263 + 11.808 + 0.9318 = \underline{7.4134}$$

$$\hat{y} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

Error

$$e = y - \text{target} = Xw + b - \text{target}$$

$$\text{where target} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$\Rightarrow e = [3.1426 - 5, 7.4134 - 6]$$

$$= \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

Gradient $\nabla_x \mathcal{L}$

$$\Rightarrow \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

$$= (1)(-1.8574) + (4)(1.4134) \quad \bigg| \quad (3)(-1.8574) + (10)(1.4134)$$

$$\Rightarrow -1.8574 + 5.6536 = \underline{3.7962} \quad \bigg| \quad -5.5722 + 14.134 = \underline{8.5618}$$

$$\nabla w = \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}$$

So we find the Bias Gradient

$$= -1.8574 + 1.4134 = \underline{-0.444}$$

$$w_3 = [-1.3316, 1.1808] - 0.01 [3.7962, 8.5618]$$

$$\Rightarrow w_3 = \begin{bmatrix} -1.3696 \\ 1.0952 \end{bmatrix}$$

$$\text{then } b_3 = 0.9318 - 0.01(-0.444)$$

$$= 0.9318 + 0.00444 = \underline{0.9362}$$

$$\text{So } w = [-1.3696, 1.0952] \quad \& \quad b = 0.9362$$